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AWS AI HACKATHON 2026

PROJECT REQUIREMENTS

Theme	AI for Real-World Impact
Focus	Build. Innovate. Solve.
Team Structure	Two members per team
Progress Deadline	**Monday - at least 80% of planned work, including architecture**

Monday deadline: at least **80% of planned work** complete, with the **architecture diagram** included and the **frontend ready for demonstration**.

1. Hackathon Theme

Each team must develop an AI-powered solution that addresses a real-world problem, preferably one that affects people or communities in South Africa. The solution should demonstrate meaningful AI usage, appropriate AWS services, and a working prototype.

2. Team Structure

- Each group must consist of **two members**.
- Both members must actively contribute to planning, development, testing, and the final presentation.
- Teams may divide responsibilities, but both members must understand the overall solution and be able to explain the project.

3. Monday Progress Deadline: 80% Completion

By **Monday, each team must be ready to present at least 80% of its planned project work**, including the proposed architecture. Teams must be prepared to demonstrate their progress and explain their implementation decisions.

The Monday presentation must include:

Item	What to cover
Problem statement	Explain the real-world problem the project aims to solve.
Proposed solution	Describe the solution, its target users, and its key features.
Architecture	Present the system architecture diagram, showing the main components, AWS services, how they connect, and how data flows through the system.
Frontend	Show a complete, functional, and demo-ready frontend.
AI integration	Demonstrate the AI functionality or show clear progress towards its implementation.
AWS implementation	Explain which AWS services are being used and why they are appropriate.
Working prototype	Demonstrate the main user journey and the features implemented so far.
Outstanding work	Identify the remaining work needed to complete the project.

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Teams should not wait until the deadline to begin preparing their architecture or integrating their components.

4. Frontend Requirements

The frontend must be **complete and ready for demonstration by Monday**. It should include:

- All essential screens and pages.
- Clear navigation and a user-friendly interface.
- Functional buttons, forms, and user interactions for the main workflow.
- A responsive layout suitable for different screen sizes.
- Integration with the backend and AI functionality where available.
- Clear labels for any simulated features that are not fully implemented.

5. Architecture and AWS Requirements

Each team must provide a **clear architecture diagram by Monday**. The architecture should identify:

- The frontend and how users access the application.
- The backend components and how requests are processed.
- The AWS services used and the purpose of each service.
- The AI service and how it contributes to the solution.
- The database or storage components, where applicable.
- The flow of information between the components.
- Relevant security and privacy considerations.

Teams should use AWS services appropriately and be able to justify their choices. The architecture must reflect the actual implementation as closely as possible, with planned or simulated components clearly identified.

6. AI Requirements

AI must serve a clear and meaningful purpose in the solution. Teams should be able to explain:

- What the AI does.
- Why AI is suitable for the problem.
- Which AWS AI service or model is being used, where applicable.
- How users interact with the AI functionality.
- The limitations of the AI and how its output is handled responsibly.

If AI functionality is simulated or not yet fully integrated, teams must clearly communicate this during the presentation.

7. Testing and Demonstration

By Monday, teams should be able to demonstrate the main user journey and test the key features that have been implemented. Teams must:

- Test the main workflow from beginning to end.
- Resolve critical errors affecting the demonstration.
- Clearly distinguish completed, simulated, and outstanding functionality.

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- Prepare a backup demonstration plan in case of technical difficulties.
- Ensure both members can explain the project, architecture, AWS services, and AI functionality.

8. Final Submission Checklist

Before presenting on Monday, each team must confirm that:

- ☐ The team consists of two members.
- ☐ At least 80% of the planned work is completed.
- ☐ The architecture diagram is ready and can be explained.
- ☐ The frontend is complete and demo ready.
- ☐ The main user journey can be demonstrated.
- ☐ The AI functionality is demonstrated, or its current progress is clearly explained.
- ☐ The AWS services and their purpose are identified.
- ☐ The implemented features have been tested.
- ☐ Completed, simulated, and outstanding features are clearly identified.
- ☐ Both team members understand the solution and can contribute to the presentation.
- ☐ The remaining work and next steps are documented.

9. Assessment Expectations

Projects will be assessed on:

- Relevance of the problem and its local impact.
- Innovation and usefulness of the proposed solution.
- Meaningful use of AI.
- Appropriate AWS architecture and service selection.
- Frontend quality and user experience.
- Prototype functionality and progress.
- Feasibility, scalability, security, and potential impact.
- Presentation quality and both members' understanding of the project.

> **Important:** Monday is a progress presentation deadline. Teams are expected to show substantial, demonstrable progress - not just an idea or a presentation. The architecture must be included, the frontend must be ready for demonstration, and at least 80% of the planned work must be completed.